Python Programming Foundations

Week 5 - Files, Errors, & Data Persistence

Day 1: "Programs That Remember"

1

Programs Can Remember

Until now, most of our programs have only worked while they were running.

Today we add persistence: the program can save information outside itself and load that information again later.

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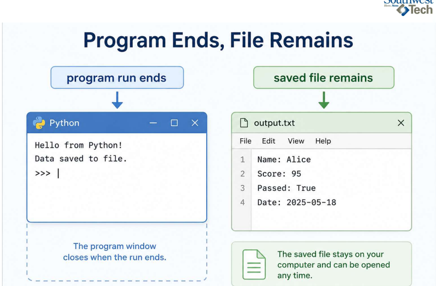

Programs Can Remember

3

Quick Review

Printing a value shows it, but it does not save it.

If the program ends and the value only lived in a variable, that value is gone.

4

The Save / Load Cycle

Persistence is a cycle:

- create or collect information
- write it to a file
- close or rerun the program
- read the file back
- display or use the loaded information

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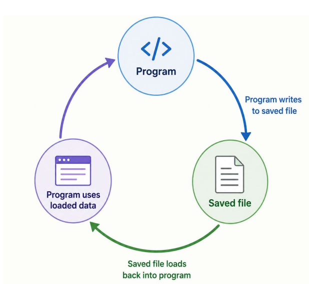

The Save-Load Cycle

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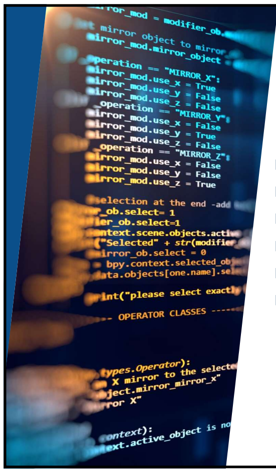

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What We Will Use Today

- a file name
- `open()`
- `with`
- write mode
- read mode
- visible output after loading

7

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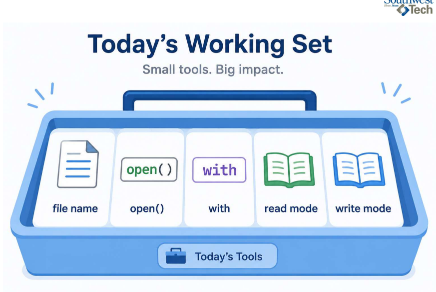

Toolbox

8

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What We Will "Skip" For Now

We will save these for later:

- databases
- binary files
- file compression
- custom JSON encoding
- complex folder and deployment issues

Today's goal is simple: save text, load text, explain the cycle.

9

Parked For Later

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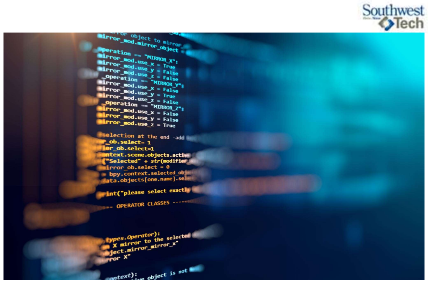

Review Coding Lab

11

What Counts as Success Today

- WRITING A FILE
- READING A FILE
- EXPLAINING THE CYCLE

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Data Persistence

13

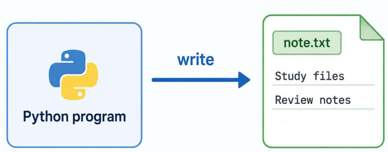
Writing Stores Data Outside The Program

Writing means the program places data into a file.

The file becomes the evidence that the program saved something outside the running program.

14

Writing Stores Data Outside the Program



The file remains after the program ends.

Storing Data

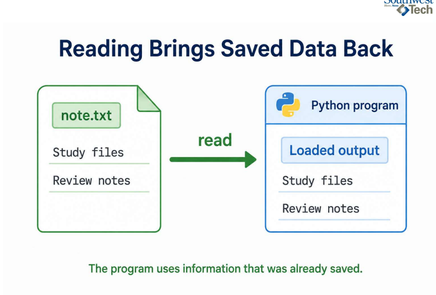
15

Reading Brings Saved Data Back

Reading means the program opens an existing file and brings its contents back into the program.

The program can then print, process, or reuse that loaded information.

16



Reading/Retrieving Data

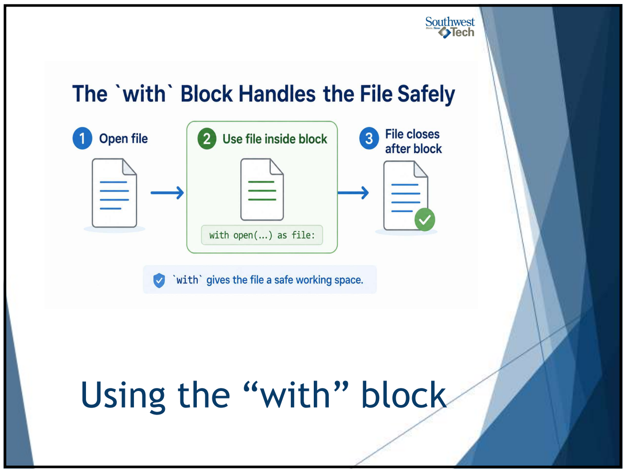
17

Why `with` Matters

`with open(...) as file:` creates a safe working block for the file.

When the block ends, Python handles the file cleanup for us.

18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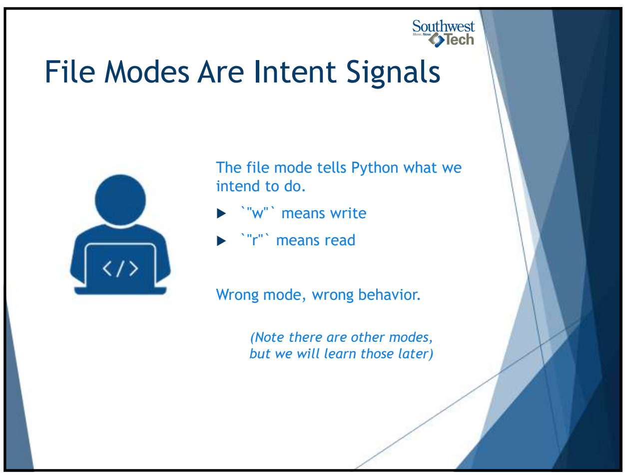
The `with` Block Handles the File Safely
1 Open file
2 Use file inside block
with open(...) as file:
3 File closes after block
`with` gives the file a safe working space.
Using the "with" block

19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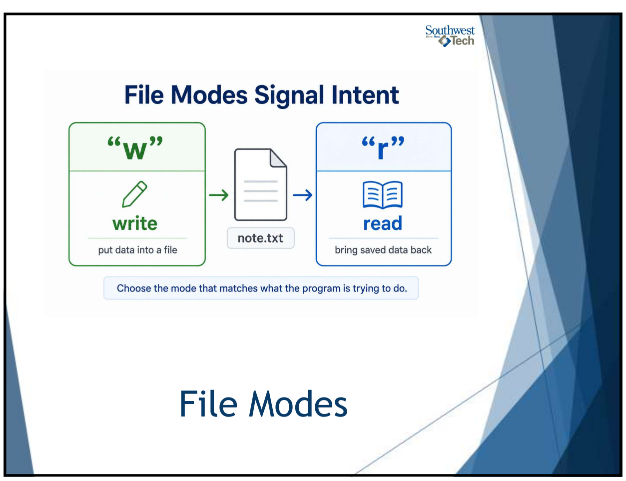
File Modes Are Intent Signals
The file mode tells Python what we intend to do.
`"w"` means write
`"r"` means read
Wrong mode, wrong behavior.
(Note there are other modes, but we will learn those later)

20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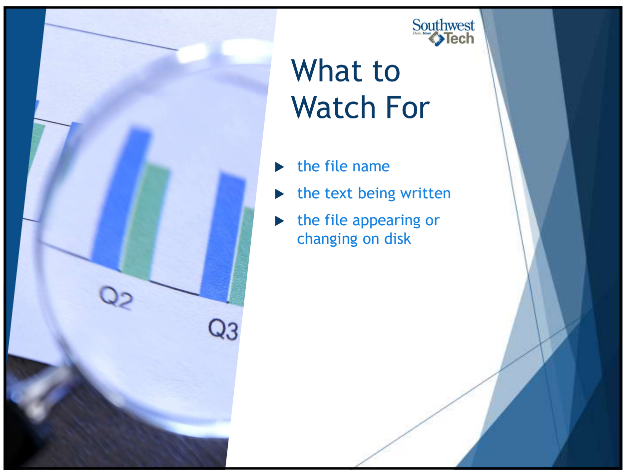
File Modes Signal Intent
"w"
write
put data into a file
note.txt
"r"
read
bring saved data back
Choose the mode that matches what the program is trying to do.
File Modes

21


Demo #1
“Write A Text File”

22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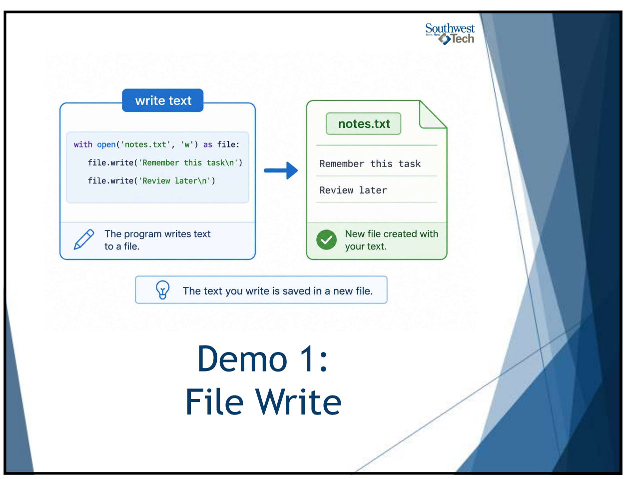
What to Watch For
the file name
the text being written
the file appearing or changing on disk
Q2
Q3

23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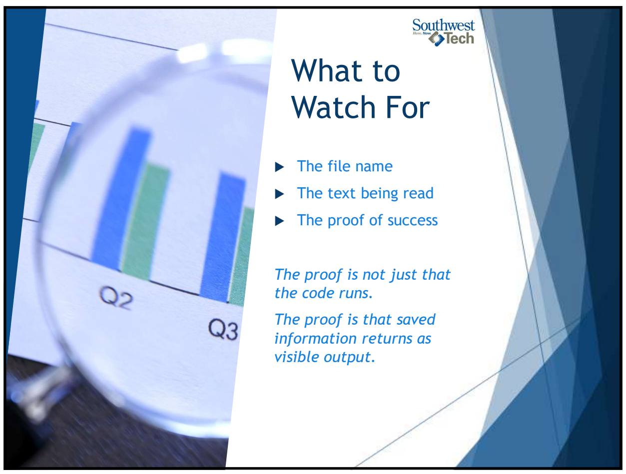
write text
with open('notes.txt', 'w') as file:
file.write('Remember this task\n')
file.write('Review later\n')
The program writes text to a file.
notes.txt
Remember this task
Review later
New file created with your text.
The text you write is saved in a new file.
Demo 1:
File Write

24


Demo #2
“Read A Text File”

25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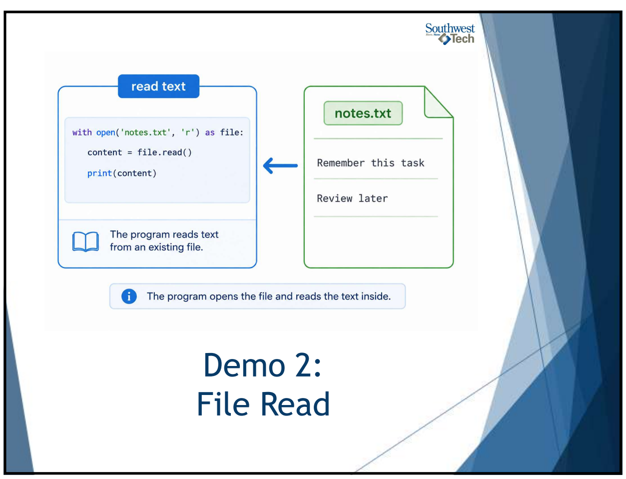
What to Watch For
The file name
The text being read
The proof of success
The proof is not just that the code runs.
The proof is that saved information returns as visible output.
Q2
Q3

26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read text
with open('notes.txt', 'r') as file:
content = file.read()
print(content)
The program reads text from an existing file.
notes.txt
Remember this task
Review later
The program opens the file and reads the text inside.
Demo 2:
File Read

27

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Common Failure: The File Is Invisible In Your Thinking

A common beginner mistake is treating the file like magic.

Instead, keep asking:

- What did I save?
- What file did it go into?
- What did I load back?

28

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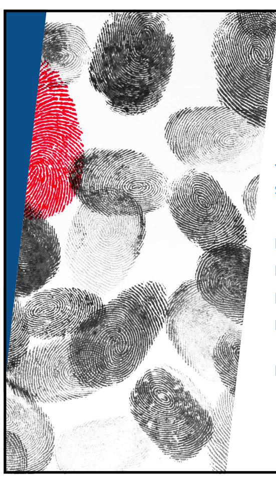
Assignment #8

In For Assignment 8, build a small save/load utility.

Possible options include

- A note keeper,
- task list
- saved preference
- tracker
- stored score.

29



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Evidence

Your submission should make the save/load behavior visible:

- Python file
- sample saved file if requested
- short README explanation
- one note about what is saved and loaded
- AI-use note if AI materially helped

30

Assignment Evidence

Assignment 8 Evidence

Include these items to show complete work.

- Python file
- saved data file
- run output
- README explanation
- AI-use note, if used

These items help show your work clearly.

31

Closing Success Check

By the end of today, you should be able to say:

"My program saved this information in this file, then loaded it back and showed it clearly."

32

Thank you!

Have Fun!

33